WATER SOLUBLE VITAMINS

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Lecture Outlines

- Introduction
- Classification
- Recommended Daily Allowance (RDA)
- Sources
- General Properties of Water Soluble Vitamins
- Individual Water Soluble Vitamins
  - Structure
  - Active forms (coenzymes)
  - Biochemical functions
  - Causes of Deficiency
    - Deficiency manifestations

What Are Vitamins?

- Tasteless **organic** compounds
- Required in **small amounts**
- **Functions**:
  - Regulate Metabolism
  - Help convert the **3 Organic Energy Yielding Nutrients** into ATP
  - Promote Growth and Reproduction

Deficiencies can result in serious illnesses.

Vitamins Discovered about 100 Years Ago

1747 - British Navy Dr. James Lind gave sailor limes to combat scurvy – **Vitamin C** deficiency.
(*Vit C not discovered until 1930's)

1897 - Dutchman Christiaan Eijkman observed chickens & people eating white rice developed Beriberi, causing heart problems & paralysis. When chickens fed unprocessed brown rice (outer **Bran** intact), no disease developed: Brown rice cured his patients too.

Connected Food to Curing Disease.

Note: Outer rice bran contains vitamin **B_1**, (Thiamin).

History of Vitamins

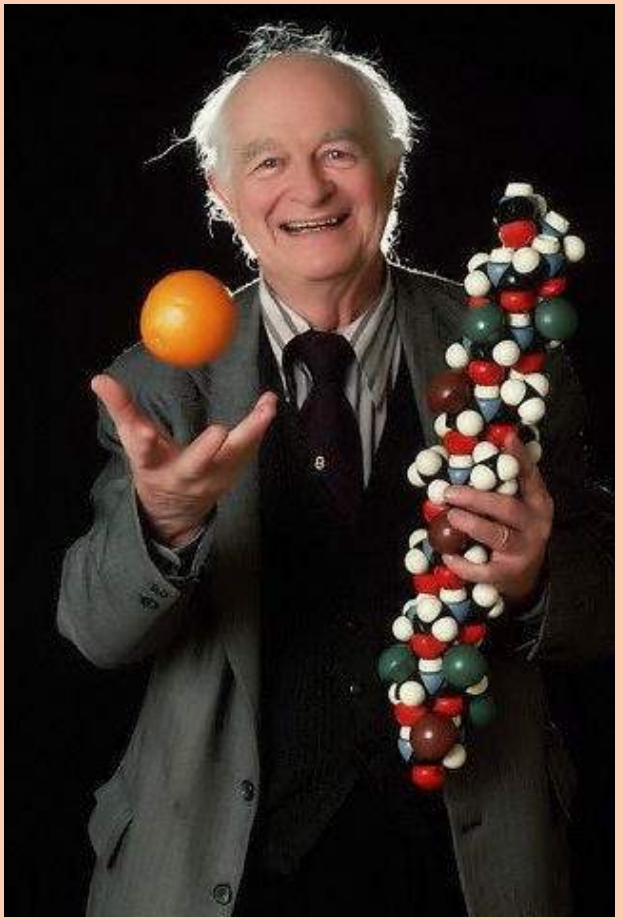
1912 - E.V. McCollum, discovered first fat soluble vitamin, **Vitamin A**; found rats fed **butter** were healthier than those fed **lard** (more Vitamin A in butter).

1912 - Dr. Casmir Funk coined term '**vitamins**' as vital factors in diet (from *vital* and *amine)*. Found these unidentified substances in food could prevent diseases of scurvy, beriberi and pellagra (from deficiency of niacin, vitamin B_3).

1940's - Water soluble B and C vitamins were identified.

1968 - Linus Pauling, created the term *Orthomolecular Nutrition*, "pertaining to the right molecule". Connecting optimum nutrition to better health.

1970's-1980's – Further studies by Pauling et al. suggested very large doses of intravenous vitamin C could help increase survival time and improve the quality of life of terminal cancer patients.



Naming Vitamins

- Each new vitamin given a temporary name when discovered.
- Names of vitamins follow alphabet, starting with A; A, B, C, D, E, and K… where we currently are
- Vitamins F, G, and H were dropped.
- Nomenclature changed when vitamin B was found to have several functions.
  - Subscripts added for each new isolated role.

The Criteria for Designating Vitamins:

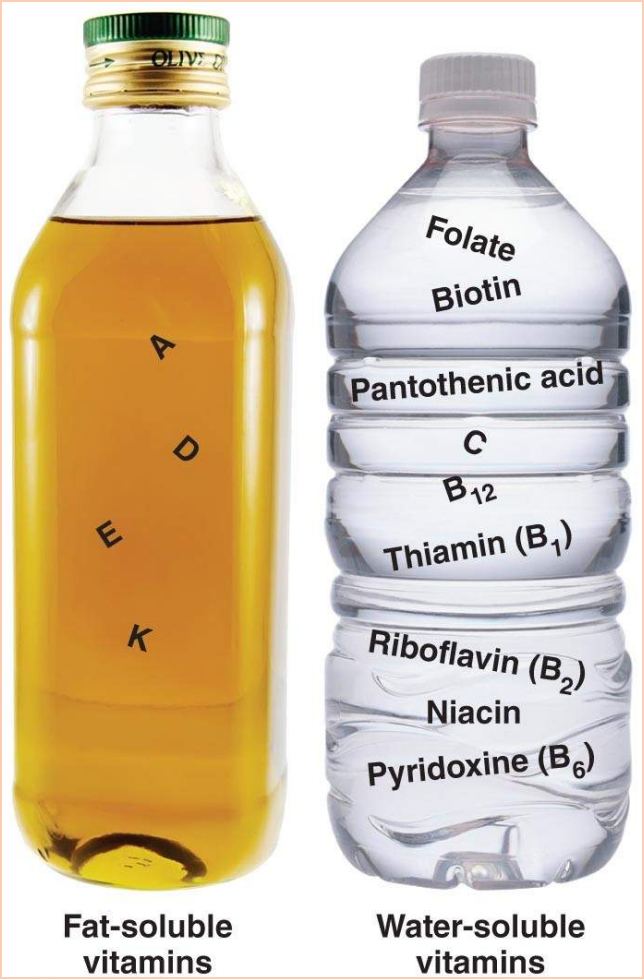
1. Cannot be synthesized in ample amounts in body
2. Chronic deficiency cause physical symptoms.
3. Symptoms disappear once vitamin level restored.
4. Deficiency can cause permanent damage.

13 compounds meet the above criteria!

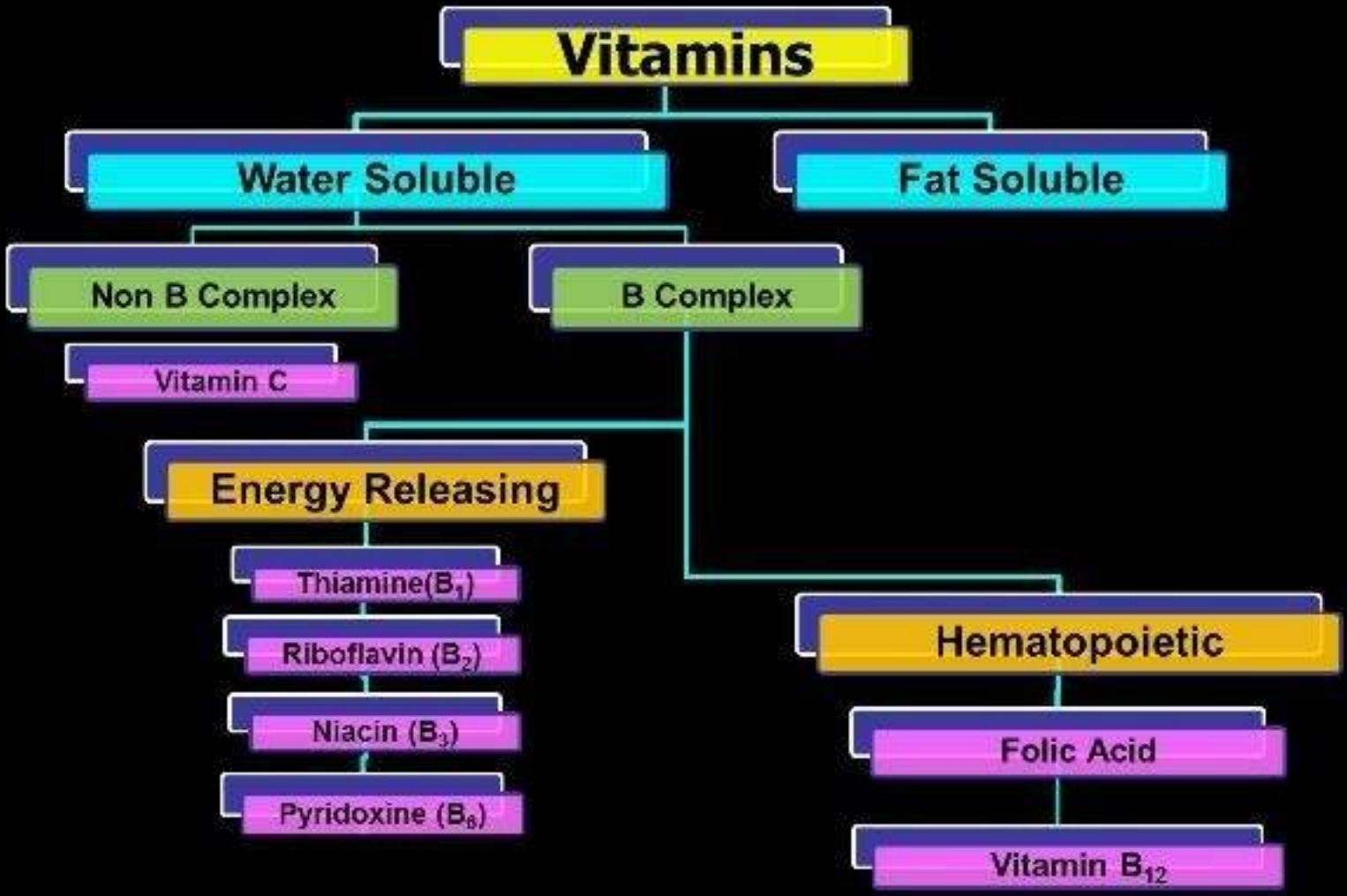
Classifying Vitamins:

Vitamins are classified based on **solubility**.

- Nine (**9**) Water-Soluble:
  - **B** vitamin complex
  - Vitamin **C**
- Four (**4**) Fat-Soluble:
  - Vitamin **A**
  - Vitamin **D**
  - Vitamin **E**
  - Vitamin **K**



Classification of Vitamins



Solubility Influences a Vitamin's…

- **Digestion**
- **Absorption**
- **Transportation**
- **Storage**
- **Excretion**



Compare Water Soluble and Fat Soluble Vitamins



Vitamins Are Organic Compounds

- All contain **C**, **H** and **O**.
- Can also contain some **N** and **S**.
- Each have a ***unique*** Chemical structure.
- Are each a singular unit.
- Are absorbed intact.
- Perform ***numerous essential functions***, some with more than one role in metabolism.

Comparison of two types of vitamin

	Fat soluble vitamins	Water soluble vitamins
Solubility in fat	Soluble	Not soluble
Water solubility	Not soluble	Soluble
Absorption	Along with lipids Requires bile salts	Absorption simple (except vitamin B_{12})
Carrier proteins	Present	No carrier proteins (except vitamin B_{12})
Storage	Stored in liver	No storage (except vitamin B_{12})
Deficiency	Manifests only when stores are depleted	Manifests rapidly as there is no storage (except vitamin B_{12})
Toxicity	Hypervitaminosis may result	Unlikely, since excess is excreated
Major vitamins	A, D, E and K	B and C

Recommended Daily Allowances (RDA)

The RDA is the average daily dietary intake level that is sufficient to meet the nutrient requirements of nearly all (97 – 98%) the individuals.

Vitamin	RDA
Thiamine (B_1)	1 to 1.5mg per day
Riboflavin (B_2)	1 – 2 mg per day
Niacin (B_3)	15 – 20 mg per day
Pyridoxine (B_6)	1.5 – 2 mg per day.
Ascorbic Acid (Vit. C)	60 – 70 mg per day
Folic Acid	200µg per day
Vitamin B_{12}	3µg per day

The RDA is not the minimal requirement for healthy individuals; rather it is set to provide a margin of safety for most individuals.

Sources	
Thiamine (B_1)	Unpolished Rice, Cereals and Whole wheat (brown) bread.
Riboflavin (B_2)	All types of food (No deficiency due to diet)
Niacin (B_3)	Cereals, nuts, milk, meat, fish (Synthesized endogenously from Tryptophan)
Pyridoxine (B_6)	Meat, Fish, Poultry, Cereals, non-citrus fruits, Potatoes etc.
Ascorbic Acid (Vit. C)	Citrus Fruits (Lemon and Oranges)
Folic Acid	Green Leafy Vegetables.
Vitamin B_{12}	Meat and Meat Products.

General Properties of Water Soluble Vitamins

- They are heterogeneous group of compounds since the differ chemically from each other.
  - The only common thing about them is their solubility in water.
- Excreted in the urine and are non-toxic to the body.
- Not stored in large quantities (except B_{12}) hence they must be continuously supplied in the diet.
- The water soluble vitamins form coenzymes (active form of the vitamin) that help enzymes to participate in a variety of biochemical reactions.

THIAMINE

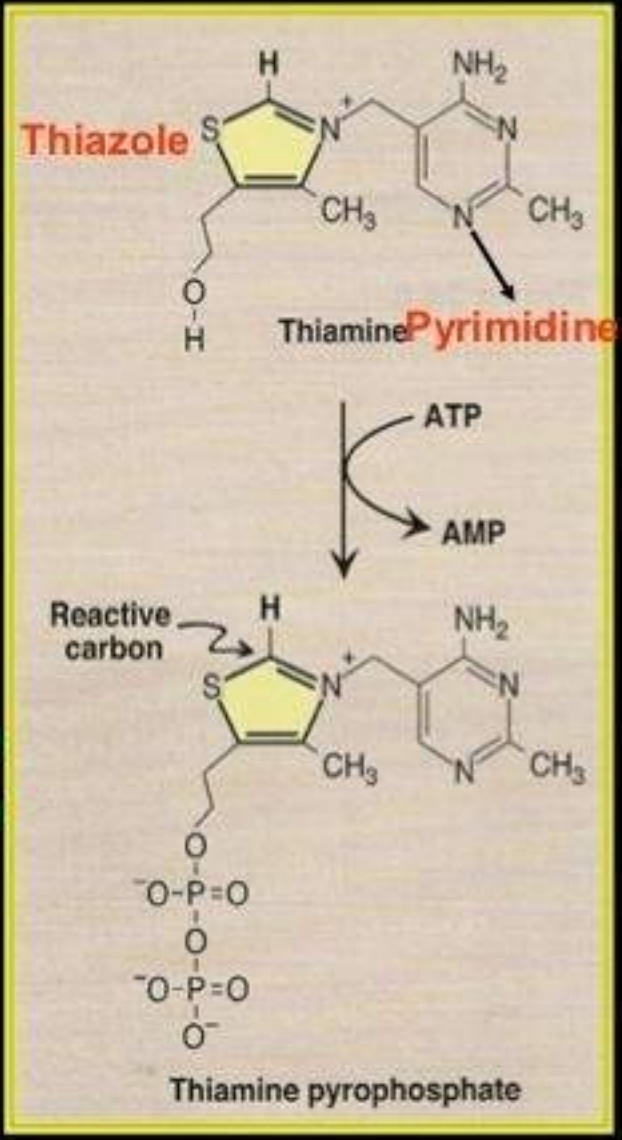
- Thiamine is also called as Vitamin B_1.

Structure And Activation

- Thiamine contains a pyrimidine ring and a thiazole ring.

- The active form of thiamine is the coenzyme Thiamine Pyrophosphate (TPP).

- Activation occurs mainly in liver.

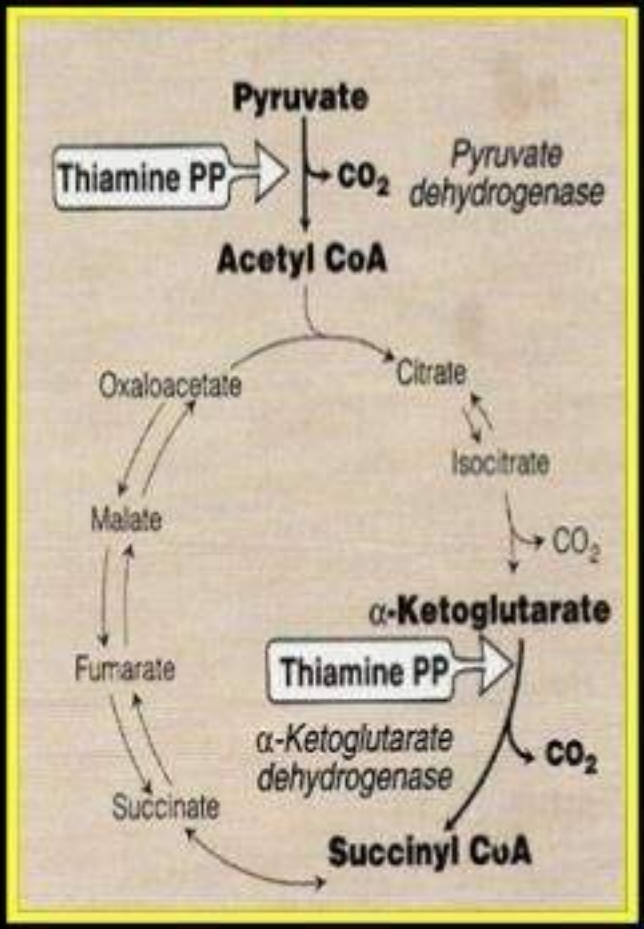


Biochemical Functions

- Thiamine pyrophosphate (TPP) is involved with the energy releasing reactions of carbohydrate metabolism.

- DECARBOXYLATION REACTIONS

1. Pyruvate Dehydrogenase (PDH)
2. α Ketoglutarate dehydrogenase.



$$\text{Pyruvate} \xrightarrow[\text{TPP}]{} \text{Acetyl CoA} + CO_2$$

(3C) (2C)

Causes of Deficiency Manifestation

- Inadequate Diet (polished rice): Thiamine is present in the outer layer of rice grains.
- Alchoholism:
- Pregnancy and lactation: Leads to increase demand.
- Presence of enzyme *thiaminase* (in raw fish) inactivates thiamine by breaking the thiazole ring.

Deficiency Manifestations of Thiamine

- Mainly seen in those consuming exclusively polished rise. Deficiency leads to a disease known as Beri-beri.
- Beri-beri is of three types.

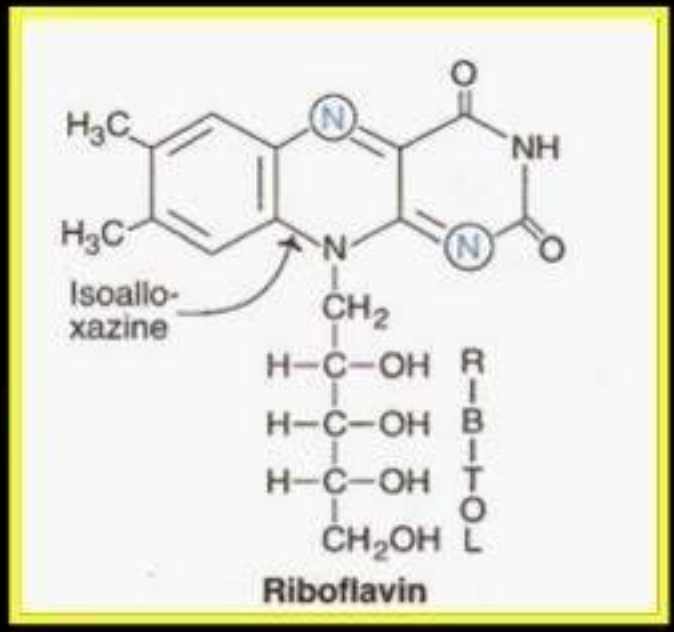
I. Wet Beri-beri: Edema and weak heart muscles

II. Dry Beri-beri: Degeneration of peripheral nerves and weak muscles.

III. Infantile Beri-beri: Affects infants.

RIBOFLAVIN

- Riboflavin is also known as Vitamin B_2



Structure:

- Isoalloxazine ring attached to Ribitol.

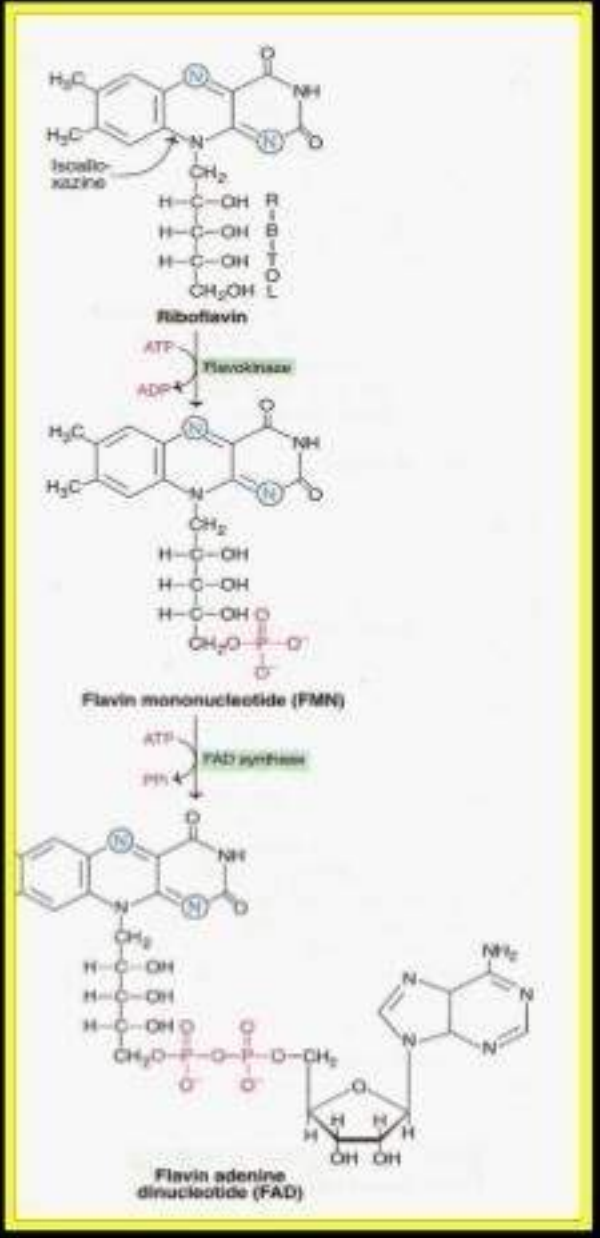
- Stable to heat but sensitive to light (photosensitive). When exposed to UV rays it gets destroyed.

Active Coenzymes of Riboflavin

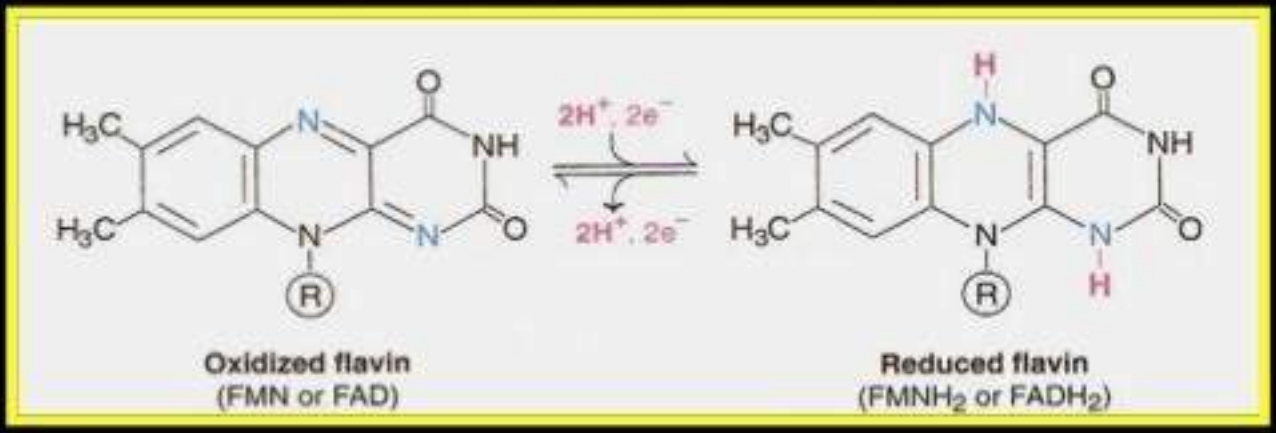
- FMN: Flavin mononucleotide
- FAD: Flavin adenine dinucleotide

FAD = FMN + AMP

Active forms are formed in the intestine and liver.



Biochemical Functions



- The isoalloxazine ring serves as an acceptor of two hydrogen atoms (with electrons)

$$FMN \longrightarrow FMNH_2$$
$$FAD \longrightarrow FADH_2$$

- The coenzymes FMN($FMNH_2$) and FAD($FADH_2$) participate in many oxidation reduction reactions and in the Electron Transport Chain.

Causes of Deficiency

- **Phototherapy:** Riboflavin being light sensitive gets destroyed.
- **Chronic Alcoholics:**
- **Pregnancy and Lactation:** There is increased demand of Riboflavin.

Deficiency Manifestation

- **Glossitis:** Smooth and purplish tongue.
- **Dermatitis:** Inflammation of the facial skin in particular.
- **Cheilosis:** Fissures at the corner of the mouth

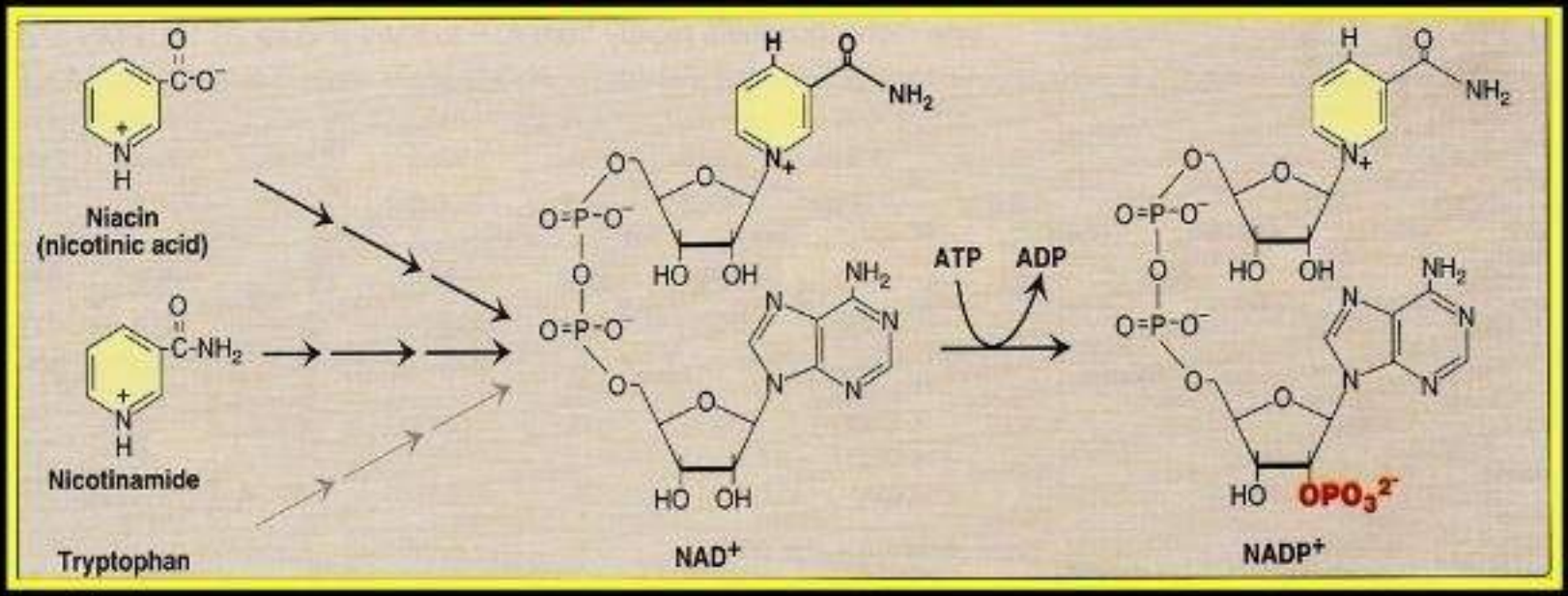


Normal



Glossitis

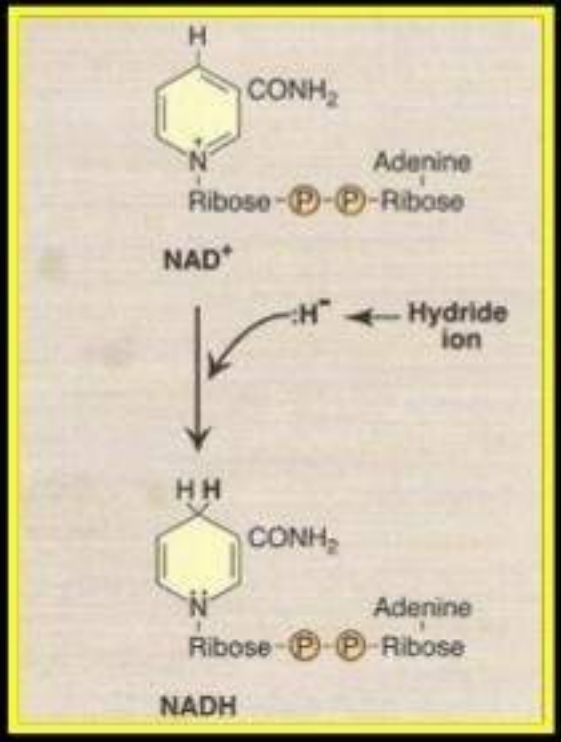
Active Coenzyme forms of Niacin

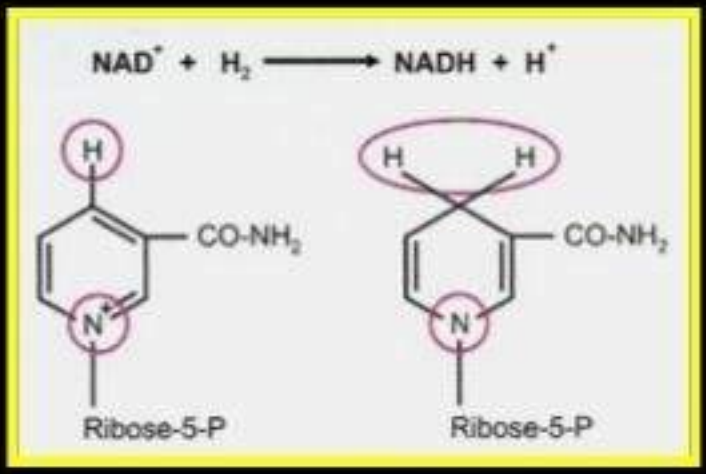


- Two active coenzymes: Formed in the Liver

NAD+ = Nicotinamide adenine dinucleotide.

NADP+ = Nicotinamide adenine dinucleotide phosphate





$2H \longrightarrow H + H^+ e^-$

- NAD$^+$ and NADP$^+$ are present in oxidized and reduced forms. They undergo reduction of the pyridine ring by accepting Hydride ion (hydrogen atom plus electron)

$$NAD^+ \xrightarrow{2H} NADH + H^+$$

$$NADP^+ \xrightarrow{2H} NADPH + H^+$$

Biochemical Functions

- The Coenzymes NAD^+(NADH) and $NADP^+$(NADPH) are involved in oxidation–reduction reactions in carbohydrate, lipid and protein metabolisms.

Therapeutic doses of Niacin 'lowers blood cholesterol' and causes vasodilatation.

Causes of Niacin Deficiency

- Inadequate Diet: Seen in people whose staple diet is maize. In maize, niacin is present but in the bound form and is unavailable.
- Pregnancy and Lactation: Because of increased demand.
- Chronic Alcoholics:
- Vitamin B_6 deficiency:

Deficiency Manifestation

- Deficiency of Niacin leads to the clinical condition called pellagra(=rough skin)

Pellagra is characterized 3 D's:

- Dermatitis: Bright red pigmentation occurs which later turns dark resulting in disfiguration of the skin.

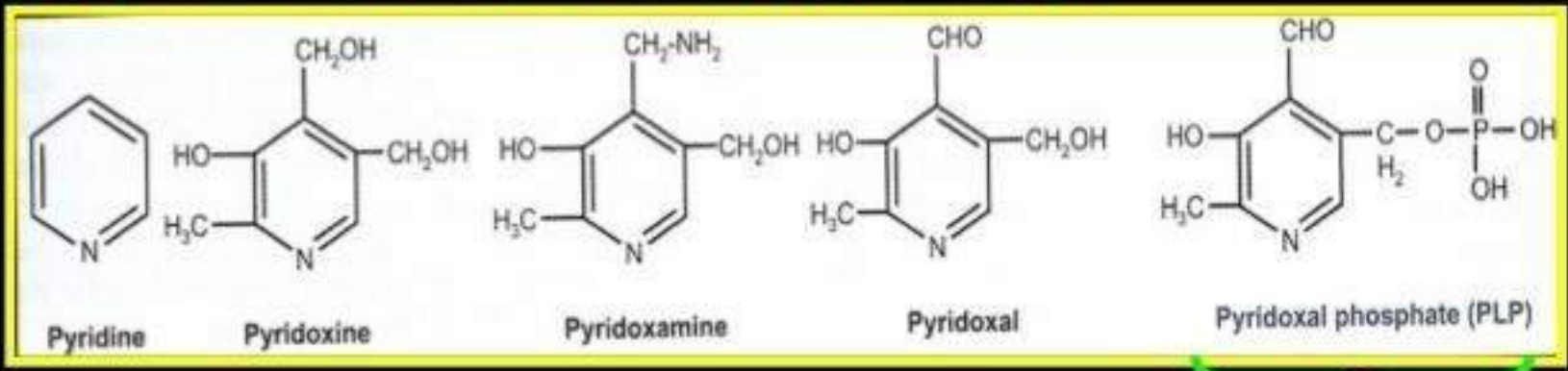


- Diarrhea: Mild to sever with blood and mucous.

- Dementia: Involvement of the nervous system. Irritability, delirium, and poor memory.

PYRIDOXINE

- Pyridoxine is also known as Vitamin B_6.



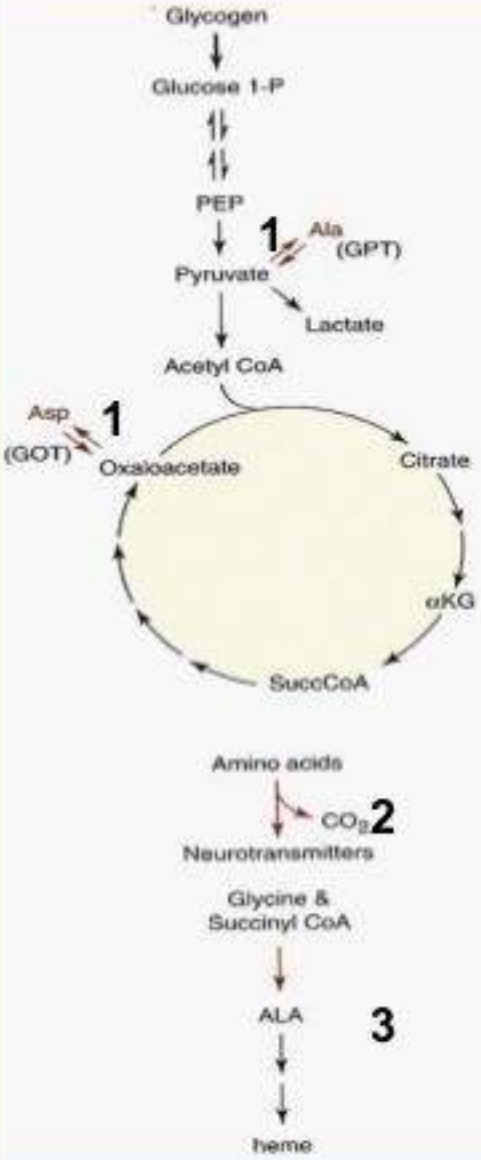
- Vitamin B_6 is used to collectively represent three compounds namely pyridoxine, pyridoxal and pyridoxamine. Structurally they are pyridine derivatives

The active Coenzyme form of B_6 is Pyridoxal Phosphate (PLP) which is synthesized in the intestine from all the three forms.

Biochemical Functions

- Pyridoxal Phosphate (PLP) the active coenzyme form of Vitamin B_6 is closely associated with the metabolism of Amino acids.

1. Transamination.
2. Decarboxylation.
3. Heme synthesis.
4. Deamination.
5. Production of Niacin



Causes of Deficiency

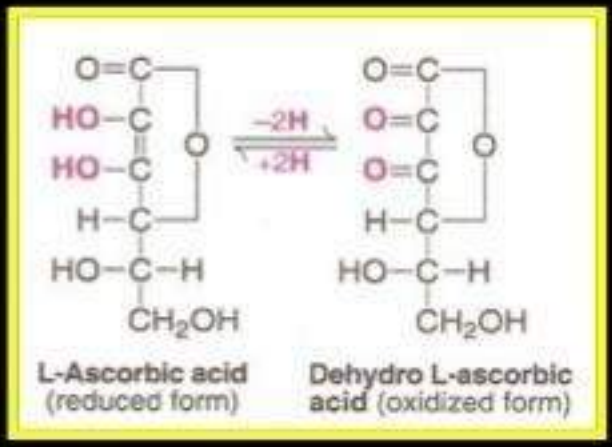
- Isoniazid: Antituberculous drug which inhibits pyridoxal phosphate formation.
- Oral Contraceptive Pills: Binds to pyridoxal phosphate and inactivate it .

Deficiency Manifestation

- Neurological Manifestation:
  Convulsions and Demylination of the nerves
- Dermatological Manifestation:
  Pellagra because Niacin is not formed from Tryptophan.
- Hematological Manifestation:
  Hypochromic Microcytic Anemia due to inhibition of Heme synthesis.

Vitamin C

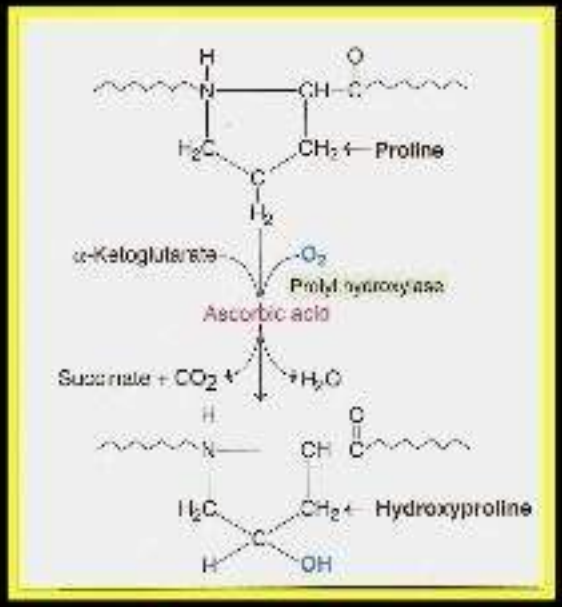
- Vitamin C is also known as Ascorbic Acid
- Structurally vitamin C resembles a carbohydrate (hexose).
- Heat sensitive and gets destroyed by cooking. In the process of cooking 70% of vitamin C is lost.



- Does not have any active coenzyme form but acts as a strong reducing agent.

Biochemical Functions

- Collagen formation:



- Vitamin C is required for hydroxylation of proline and lysine. The hydroxyproline and hydroxylysine formed are essential for the collagen cross-linking and the strength of the fibre.

- **Antioxidant Role:** Vitamin C acts as an antioxidant preventing tissue injury due to oxidative damage by free radicals:
  - Fights infection.
  - Reduces the risk of cancer and coronary artery disease.

- **Immunological Function:** Vitamin C enhances the synthesis of Immunoglobulins (antibodies) and increases the phagocytic functions of leukocytes.

Deficiency Manifestations

Deficiency is manifested as Scurvy which is characterized by:

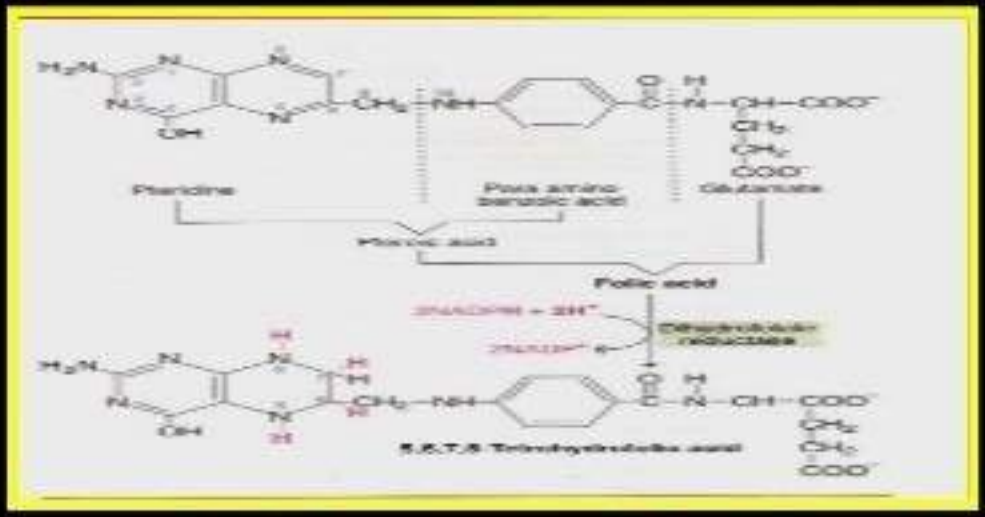
- Hemorrhages under the skin, bone fragility, joint pain.
- Poor wound healing, frequent infections.
- Spongy and bleeding gums, loosened teeth.



FOLIC ACID

- Folic Acid Deficiency is the most common vitamin deficiency particularly among pregnant female.

Structure and Active Form

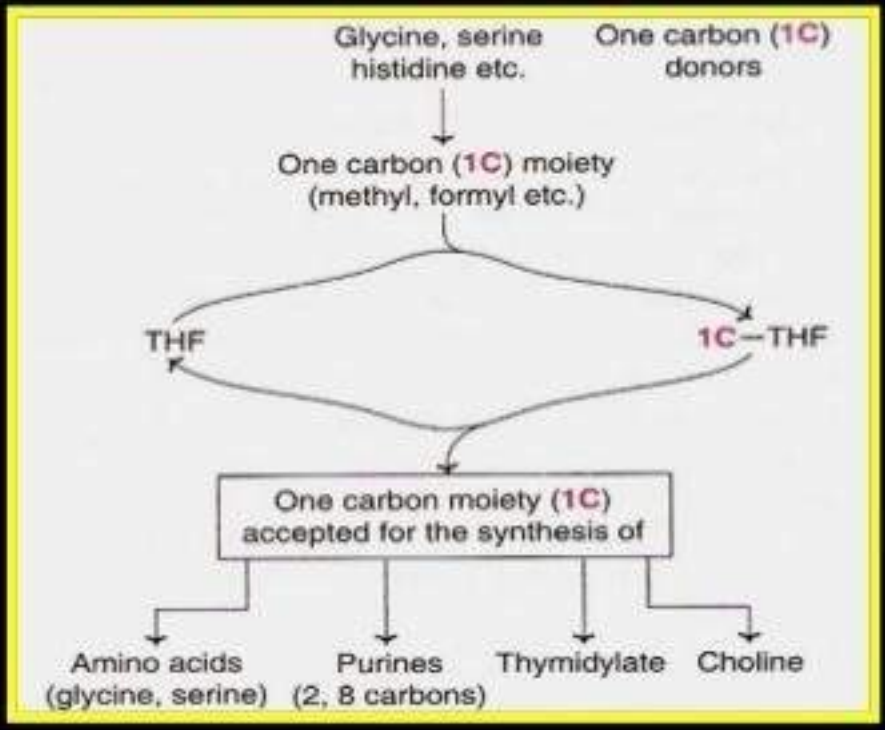


- It is composed of three constituents. Pteridine linked with para-aminobenzoic acid (PABA) is called Pteroic acid.

- It is then attached to glutamate to form Folic Acid. Active form is Tetrahydrofolic Acid (THF)

Biochemical Functions

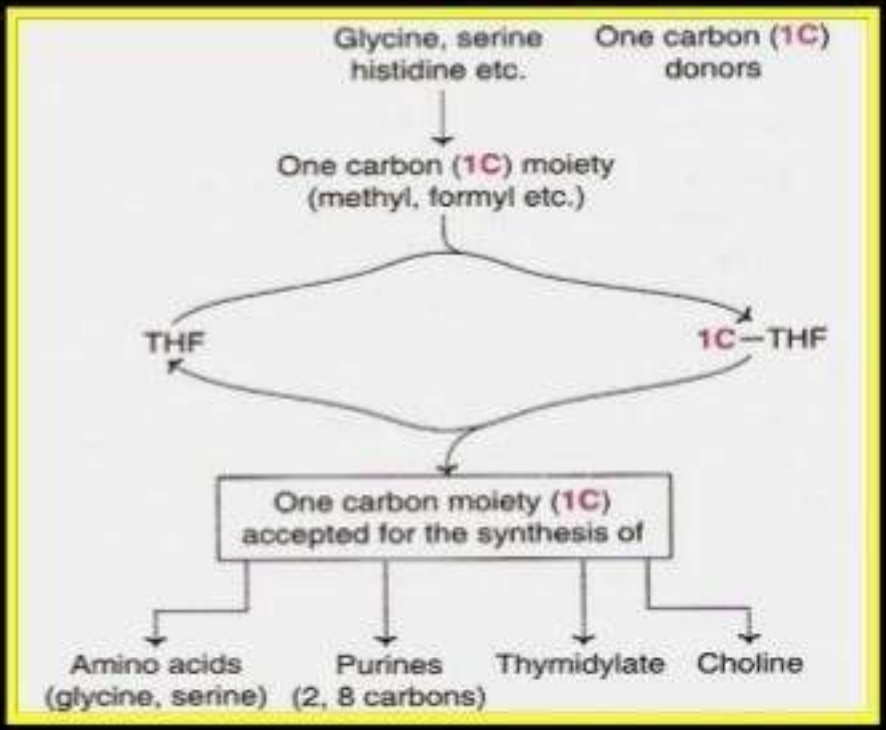
- The active coenzyme tetrahydrofolate (THF) is involved in One Carbon Metabolism.
- THF serves as acceptor or donor of one carbon units (formyl, methyl etc.) in a variety of reactions involving amino acid and nucleotide metabolism.



- One carbon compound is an organic molecule that contains only a single carbon

  - Methyl (- CH_3)
  - Methenyl (- CH=)
  - Methylene (- CH_2-)
  - Formyl (- CHO)

Form Different analogs of THF



- C_2 and C_8 of Purine ⟶ Incorporated into DNA

- Synthesis of choline: For phospholipids and Acetylcholine

- Homocysteine —(B_{12})→ Methionine

(N_5MethylTHF → THF)

Causes of Deficiency

- Pregnancy: Increased requirement because of fast cell division.
- Diet: Absence of vegetables in the diet.
- Drugs: Methotrexate (anti-cancer drug) inhibits Dihydrofolate reductase so no active THF formed.
- B_{12} deficiency

Deficiency Manifestation

- Macrocytic Anemia (large distorted RBC with immature nucleus) with megaloblasts in bone marrow.
- Neural Tube Defects: Folic acid deficiency during early pregnancy may lead to neural tube defects (Spina Bifida) in the fetus.

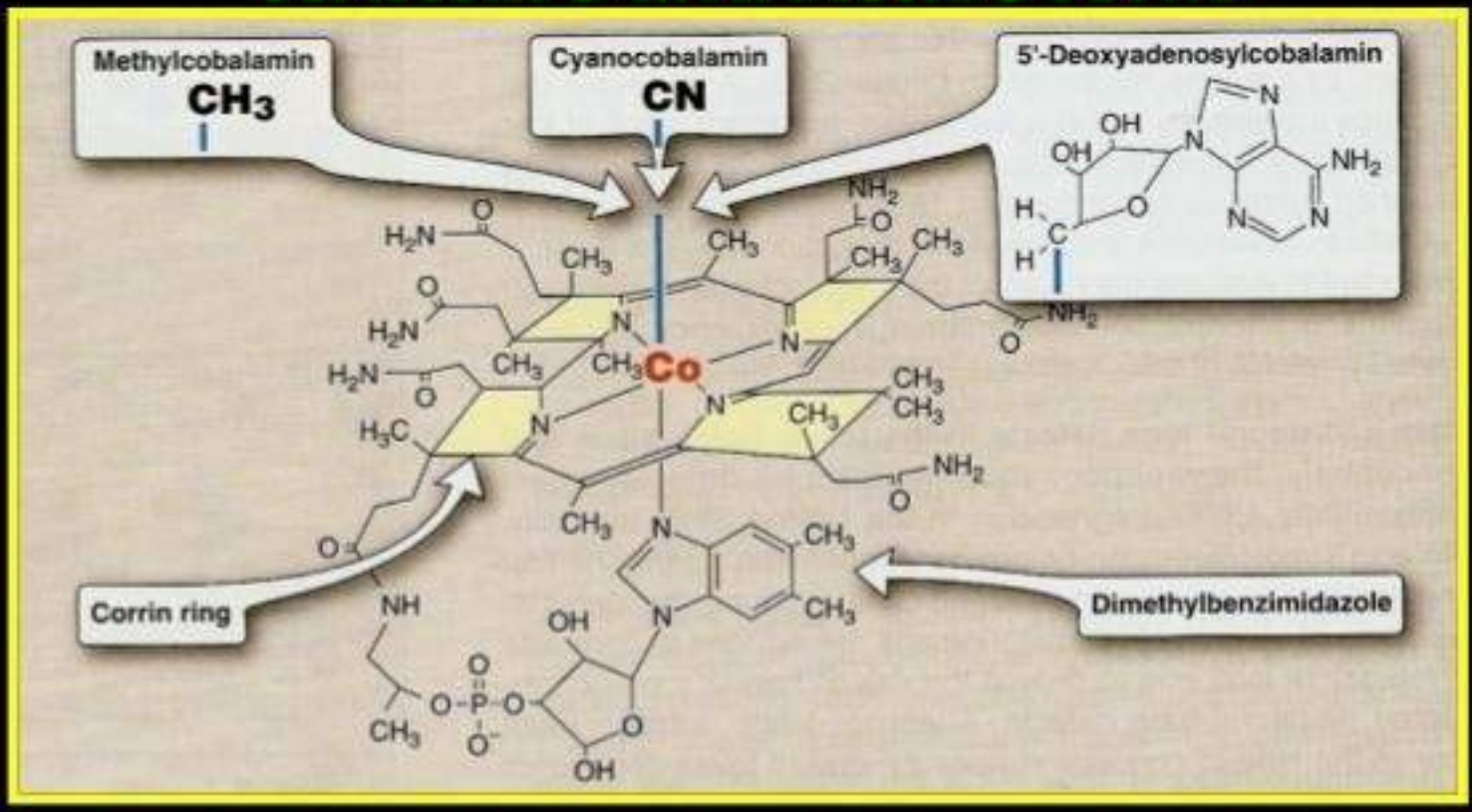


The hematological picture resembles Vitamin B_{12} deficiency.

Vitamin B_{12}

- Vitamin B_{12} is also known as Cobalamin

Structure and Active form



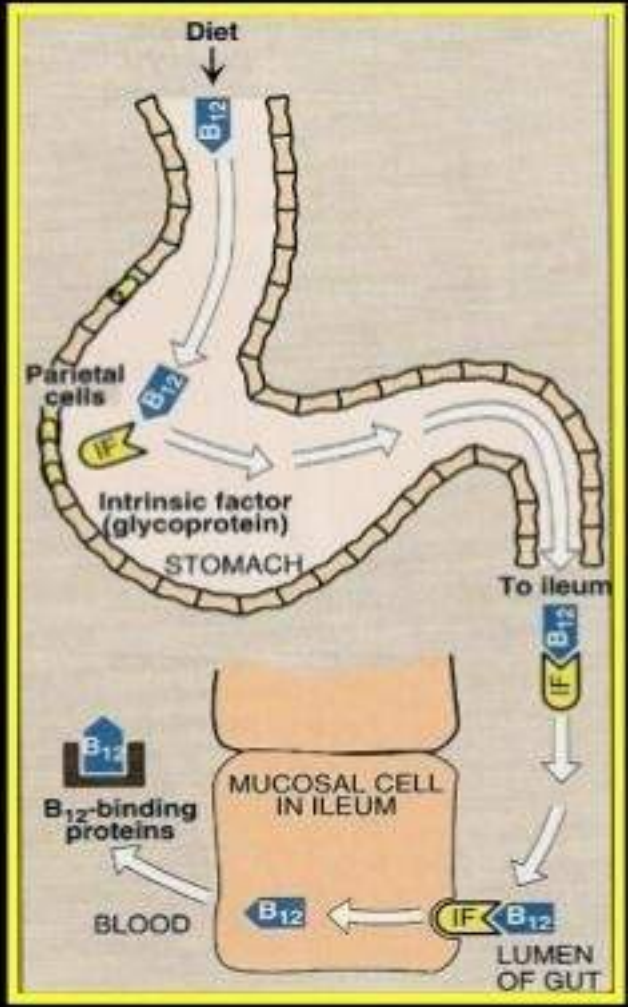
- Cobalamin contains a corrin ring with four pyrolle rings and COBALT at the centre.

- There are two active coenzyme form:
  - Deoxadenosyl Cobalamin:
  - Methylcobalamin

Absorption and Storage

- Absorption of Vitamin B_{12} requires a glycoprotein known as intrinsic factor produced in the stomach in presence of HCL.

- Stored in the liver in the form of Deoxyadenosyl Cobalamin.



Biochemical Functions

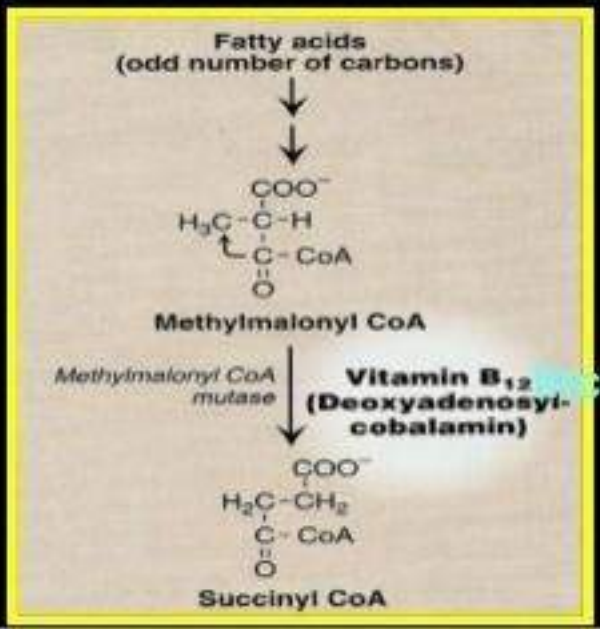
1. Synthesis of Methionine from Homocysteine

- Homocysteine —(B_{12})→ Methionine

(N_5MethylTHF → THF)

- Thus B_{12} deficiency results in decreased THF that leads to reduced nucleotide and DNA synthesis.

2. Isomerization of Methyl malonyl CoA:



Vitamin B_{12} is Deficient

↓

↑ Abnormal Fatty Acids

↓

orporated into cell membrane
(nervous system)

↓

Neurologic manifestations

Causes of Deficiency

- Vegetarians: Vitamin B_{12} is mainly present in animal diet.
- Decreased Absorption: Due to Gastrectomy and malabsorption diseases.
- Pernicious Anemia: Antibodies are formed against Intrinsic Factor leading to decreased absorption of Vitamin B_{12}.
- Pregnancy, lactation and alcoholics:

Deficiency Manifestation

- Megaloblastic Anemia: The peripheral blood shows Megaloblasts with large immature nucleated RBC.
- Nervous Manifestation: Degeneration of nervous system due to demyelination.

Megaloblastic anemia should always be treated with both folic acid and vitamin B_{12}.

THANK YOU